

When Reverse Reachability Has No Domain: Influence Maximization under Threshold-Dependent Overexposure

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Abstract. Influence maximization (IM) is dominated by reverse-reachability (RR) sampling, which reduces the problem to a maximum-coverage instance and yields near-linear-time algorithms with $(1 - 1/e - \varepsilon)$ guarantees. That reduction rests on two assumptions: a node can become active after receiving influence from a single successful edge, and the influence function is monotone submodular. We study targeted IM under a *threshold-dependent overexposure* model in which a node becomes positively active only when the total weighted influence it receives from its in-neighbours falls inside a node-specific window $[\theta_v^\kappa, \theta_v^\tau]$. We prove that RR sampling is not merely slow here but *structurally inapplicable*: because the window is drawn uniformly from the simplex $\{0 \leq \theta^\kappa \leq \theta^\tau \leq 1\}$, $\theta_v^\kappa = 0$ is a null event, so a node with zero incoming influence can never be positive and the reverse walk has no base case. Empirically every RR set is empty (0/15233 nodes admit a window containing zero; the TIM-style estimate is 0.0 against a true spread of 111.9). Independently, the exposure score is a *set function* of the ever-activated in-neighbours, so given the windows it is deterministic in the seed set and neither a live-edge graph nor a trigger set can be sampled. We then characterise the regime in which this matters. Across nine graphs spanning mean degree 4.2 to 56.0, sweeping the seed *fraction* $|S|/n$ rather than the absolute budget, out-degree turns into an *anti-signal*: its rank correlation with the true marginal gain is positive only on the sparsest graph and negative on the other eight once $|S|/n \geq 20\%$, where 8% to 35% of candidates have a negative marginal gain. Exact Monte-Carlo greedy also degrades there, because it searches for the largest marginal with noisy estimates while the true marginal is near zero; a training-free two-hop closed form beats it, and every other baseline we tested, at every saturated fraction, and wins on 8/8 independent selection draws. We close with a negative result that delimits the remaining gap: a learned state-conditioned scorer does *not* beat that closed form on held-out states.

Keywords: Influence maximization · Overexposure · Reverse reachability · Threshold models · Negative results

1 Introduction

Influence maximization (IM) asks for a seed set of size k that maximises expected influence spread. The dominant algorithmic family is reverse-reachability (RR)

sampling [1, 11, 9]: sample random reverse walks, observe that $\mathbb{E}[\sigma(S)] = n \cdot \Pr[S \cap \mathcal{R} \neq \emptyset]$, and solve a maximum-coverage problem on the sampled sets. RR is attractive because it is both fast in practice and provably near-optimal. Its guarantees, however, rest on a specific generative structure: an independent per-edge random mechanism (the live-edge graph of independent cascade, or the trigger set of linear thresholds) together with a monotone submodular spread function.

This paper is about a setting where that structure is absent, and about what a practitioner should do instead. We study *targeted* IM under a threshold-dependent overexposure model [13] motivated by advertising fatigue: a user becomes positively active only when the weighted influence they receive from their in-neighbours lands inside a personal window $[\theta_v^\kappa, \theta_v^\tau]$. Too little influence leaves them inactive; too much makes them *negatively* active, and a negatively active user never promotes the item. The window is drawn uniformly from the simplex $\{0 \leq \theta^\kappa \leq \theta^\tau \leq 1\}$, and the exposure a node accumulates is the sum of the weights of the in-neighbours that were *ever* positively activated.

Our first contribution is negative and, we would argue, the most useful part of the paper: *RR sampling has no domain here*. Two independent obstructions apply. First, because $\theta_v^\kappa = 0$ is a null event, a node receiving no influence can never be positive, so the reverse walk that RR sampling is built on has no base case — every RR set is empty. Second, and more fundamentally, the exposure score is a set function of the ever-activated in-neighbours, so once the windows are fixed it is *deterministic* in the seed set; the only randomness in the whole process is the window draw itself. There is therefore no per-edge random structure to sample. The classical identity does not merely lose accuracy, it degenerates: we measure a TIM-style estimate of 0.0 where the true spread is 111.9. This is consistent with the source model’s own analysis, which guarantees only a γ/k ratio and optimises a monotone surrogate $\lambda(S) \geq \sigma(S)$ rather than the objective itself.

Our second contribution is to identify *when* this matters. A naive experiment measures out-degree against the true marginal gain and concludes that degree is fine — we made exactly that mistake, twice, before locating the flaw. The controlling variable is the seed *fraction* $|S|/n$, not the absolute budget. On a large sparse graph, a budget of 200 seeds is 1.3% coverage and nothing interesting happens; on a small dense graph the same budget is 42% coverage and the objective becomes genuinely non-monotone. Sweeping the fraction across nine graphs with mean degree from 4.2 to 56.0, we find that out-degree becomes an *anti-signal*: it is positively correlated with the true marginal gain only on the sparsest graph, negative on the other eight when $|S|/n \geq 20\%$, and 8% to 35% of candidates have a negative marginal gain there.

Third, we show that the standard exact reference also fails in that regime. Monte-Carlo greedy searches for the largest marginal gain using noisy estimates; when the true marginal is near zero, noise drives its decisions. On Congress-Twitter, adding the highest-degree ten nodes *reduces* spread by 5.17 ± 0.40 while

a training-free two-hop closed form adds $+2.26 \pm 0.24$ and exact greedy adds only $+0.42 \pm 0.31$. The closed form wins on 8/8 independent selection draws.

Finally, we report a negative result that bounds how much room is left. Having established that a closed form beats degree and greedy, we asked whether a learned state-conditioned scorer can beat the closed form. Using leave-one-context-out evaluation over 480 labelled states, it cannot: the closed form’s regret against the true top-5 is 0.34–0.50, and every learned variant we tried is worse, including one that wins 0/12 held-out contexts. This is a limitation of the learned approach at the tested scale, not a proof of impossibility, and we state it as such.

Contributions.

- We prove and empirically verify that reverse-reachability sampling is structurally inapplicable under threshold-dependent overexposure, for two independent reasons (§4). The result does not depend on non-monotonicity.
- We identify the seed *fraction* as the controlling variable, and show that out-degree degrades to an anti-signal across nine graphs once the network saturates (§5).
- We show that exact Monte-Carlo greedy also fails in that regime, because it is noise-driven exactly where the marginal is near zero, and that a training-free closed form dominates it — and every other baseline the source model reports against (§7).
- We report a measurement caveat that constrains the whole area: in the negative-marginal regime the label signal-to-noise ratio is below 1.4 at 60 Monte-Carlo trials, so rank-correlation claims below 300 trials are not interpretable (§7).
- We delimit the remaining gap with a negative result on learned scorers (§8).

2 Related Work

Scalable influence maximization. The reverse-reachability family underpins the fastest known IM algorithms. TIM and TIM+ [1, 12] and IMM [11] sample reverse walks and solve maximum coverage, achieving $(1 - 1/e - \epsilon)$ guarantees with near-linear expected running time; OPIM [10] extends this to the online setting; [9] tightens the stopping rule. GPU-parallel implementations [5, 8] report speedups of two to three orders of magnitude over sequential baselines. All of these inherit the identity $\mathbb{E}[\sigma(S)] = n \cdot \Pr[S \cap \mathcal{R} \neq \emptyset]$, and therefore all of them require the graph-structured randomness that §4 shows is absent here. Our result is not that these algorithms are slow, but that they cannot be instantiated.

Learning-based influence maximization. A complementary line replaces the influence oracle with a learned predictor, either to prune the candidate set before exact greedy [3] or to estimate marginal gains directly. A recent benchmark study [6] finds that, across a range of IM and maximum-coverage tasks, classical lazy greedy and sampling-based methods remain competitive with or superior to deep reinforcement learning approaches. Our §8 reaches a consistent conclusion

in the overexposure setting, via a different route: the training-free closed form is not beaten by the learned scorers we tested.

Overexposure and negative influence. That excessive exposure can be counter-productive is well documented. The model we study is due to [13], who introduce the threshold window and derive the $2\delta(1 - \delta)$ activation probability; they establish non-monotonicity and non-submodularity of the objective and propose an incremental greedy algorithm with a γ/k ratio plus an upper-bound method built on two monotone submodular linear-threshold surrogates. An earlier line models overexposure through a logistic opinion-formation term inside a targeted-IM objective [7], and notes explicitly that because the spread function is non-submodular, the greedy step cannot be accelerated with lazy evaluation or reverse-influence sets. Our contribution relative to this line is to give a structural reason — rather than an empirical observation — for why reverse-influence machinery does not apply, and to characterise the seed-fraction regime in which the resulting failure of degree-based heuristics becomes visible.

Threshold and complex-contagion models. IM has also been studied under threshold dynamics in which adoption requires reinforcement from multiple neighbours. Reinforcement learning has been applied to such settings [2]. Our finding adds a boundary condition to this literature: as soon as the activation rule depends on an accumulated *sum* that must land inside an interval, the per-edge random structure that sampling-based methods rely on is lost, and the guarantees that accompany them do not transfer.

CELF, and the cost of a violated precondition. CELF [4] is the standard lazy-evaluation accelerator for greedy maximisation and is cited as a baseline by the source model. Its saving comes from a stale-upper-bound argument: if the recomputed gain of the heap head still dominates the runner-up’s stale gain, the head is the true argmax and the remaining candidates can be skipped. That argument requires the objective to be monotone and submodular. The objective here is neither — the source model’s Theorem 2 establishes non-monotonicity and its Theorem 3 establishes that neither submodularity nor supermodularity holds — so a candidate whose gain has fallen can rise again later and the skip test is unsound. We therefore implement CELF as the *exact* greedy it is designed to accelerate; a lazy variant would yield an artificially weaker baseline whose shortfall came from a violated precondition rather than from the algorithm. We raise the point because it generalises: every overexposure objective inherits the same obstruction, and lazy evaluation is one of the few standard accelerations that is *not* blocked by the absence of reverse reachability, so it is worth knowing that it fails for a different reason.

3 Preliminaries

3.1 Targeted influence maximization

Let $G = (V, E)$ be a directed graph with $|V| = n$, and let $\omega_{uv} \in [0, 1]$ be the influence weight of edge (u, v) . A *target set* $T \subseteq V$ is given, and the goal is to choose a seed set $S \subseteq V$ with $|S| \leq k$ maximising the expected number of positively activated nodes in T . Write $\sigma(S)$ for that expectation.

3.2 Threshold-dependent overexposure

Each node v carries a *threshold window* $[\theta_v^\kappa, \theta_v^\tau]$. The pair is drawn uniformly from the $2D$ simplex

$$\Delta = \{(\theta^\kappa, \theta^\tau) \in [0, 1]^2 : \theta^\kappa \leq \theta^\tau\}, \quad (1)$$

that is, with density 2 on Δ . Here θ_v^κ is an activation threshold and θ_v^τ an overexposure threshold.

Write $A_t^{\text{in}}(v)$ for the set of in-neighbours of v that were positively activated at some time $\leq t$. Crucially, this set *keeps* nodes that were later turned negative — their influence is assumed to persist. The *exposure score* of v at time t is

$$\delta(v, t) = \sum_{u \in A_t^{\text{in}}(v)} \omega_{uv}. \quad (2)$$

States are {inactive, positive, negative}. At $t = 0$ the seeds are positive and remain so. At $t + 1$ the exposure of the neighbours of the nodes that turned positive at t is updated, and each such node transitions by

$$v \text{ becomes } \begin{cases} \text{positive} & \text{if } \theta_v^\kappa \leq \delta(v, t+1) \leq \theta_v^\tau, \\ \text{negative} & \text{if } \delta(v, t+1) > \theta_v^\tau, \\ \text{inactive} & \text{otherwise (re-evaluated when more influence arrives).} \end{cases} \quad (3)$$

Transitions are one-way: inactive $\rightarrow$ positive $\rightarrow$ negative, never backwards. A negatively active node does not promote. We call this the *threshold-window* process.

Remark 1 (Activation probability). Integrating over the window draw gives the marginal positive-activation probability as a function of exposure alone,

$$\Pr[\theta^\kappa \leq \delta \leq \theta^\tau] = 2\delta(1 - \delta), \quad (4)$$

which we denote $g(\delta)$. This is the quantity the source model builds its analysis on. It is 0 at $\delta \in \{0, 1\}$ and peaks at $\delta = 1/2$.

Two structural facts drive everything that follows and are worth isolating.

Observation 1 (Exposure is monotone in time, the probability is not). $A_t^{\text{in}}(v)$ grows with t and never shrinks, because a node that has been positively activated keeps contributing even after turning negative. Hence $\delta(v, t)$ is non-decreasing in t for every v . The activation probability $g(\delta) = 2\delta(1-\delta)$ is therefore non-monotone in time: it rises while $\delta < 1/2$ and falls once $\delta > 1/2$. The only source of non-monotonicity in the model is the shape of g , not the exposure accumulation.

Observation 2 (Exposure is a set function of the seed set). Condition on a fixed draw of all windows. Then for each v the indicator of “ v was ever positively activated” is a deterministic function of S , and consequently so is $\delta(v, t)$ for every v and t , and so is the final spread. The only randomness in the entire process is the single window draw.

Observation 2 is the reason the usual sampling machinery has nothing to work with, and we make that precise next.

3.3 Reverse-reachability sampling, and what it requires

RR sampling relies on the identity

$$\mathbb{E}[\sigma(S)] = n \cdot \Pr[S \cap \mathcal{R} \neq \emptyset], \quad (5)$$

where $\mathcal{R}$ is a random set generated by a reverse walk from a uniformly random node. The identity is available whenever the process admits a *randomised graph structure* over edges — the live-edge graph in independent cascade, the trigger set in linear thresholds — such that v is active iff it is reachable from S in that structure. Two features are required: (i) reachability can start from a base case in which a node is active under *incoming influence from a single edge or a single predecessor*, and (ii) the structure is sampled independently of S .

4 Reverse Reachability Has No Domain Here

We give two independent obstructions. The first is a base-case failure and is easy to verify empirically; the second is more fundamental and rules out any repair of the reduction.

4.1 The reverse walk has no base case

Proposition 1 (Null activation threshold). Let $(\theta_v^\kappa, \theta_v^\tau)$ be drawn with density 2 on the simplex Δ of Eq. equation 1. Then $\Pr[\theta_v^\kappa = 0] = 0$, and consequently a node with $\delta(v, t) = 0$ is positively activated with probability $g(0) = 0$.

Proof. $\{\theta^\kappa = 0\}$ is the lower-dimensional face of Δ and has measure zero under a density on the two-dimensional simplex. The second statement is Remark 1: $g(0) = 2 \cdot 0 \cdot 1 = 0$.

Table 1. Reverse reachability on NetHEPT. Every RR set is empty, so the classical estimator degenerates to zero.

quantity	value
nodes whose window contains 0	0 / 15233
RR-set size (min / median / mean / max)	0 / 0 / 0.0 / 0
RR-set size as fraction of the graph	0.0000
TIM-style estimate $n \cdot \Pr[S \cap \mathcal{R} \neq \emptyset]$, $ S = 50$	0.0
true Monte-Carlo spread $\sigma(S)$, $ S = 50$	111.93

Now consider the reverse walk used to build $\mathcal{R}$. Such a walk traverses edges backwards from a source and admits a predecessor u into $\mathcal{R}$ when u *could be active given that it has received no influence*. Under Eq. equation 3 this requires $\theta_u^\kappa \leq 0 \leq \theta_u^\tau$, hence $\theta_u^\kappa = 0$. By Proposition 1 that event has probability zero, so no predecessor is ever admitted and the walk terminates immediately.

Corollary 1 (Empty RR sets). *For a non-seed source, $\mathcal{R} = \emptyset$ almost surely. The identity Eq. equation 5 therefore returns 0 for every seed set, against a true spread that is generally positive.*

Empirical confirmation. Table 1 reports the measurement on NetHEPT ($n = 15233$, $m = 32235$, in-weights normalised so that $\sum_u \omega_{uv} = 1$ for every v with in-edges, matching the source model’s setting). We ran 300 reverse breadth-first walks from uniformly random sources, where a predecessor is admitted exactly when $0 \in [\theta_u^\kappa, \theta_u^\tau]$.

4.2 The reduction has no randomised structure to sample

The base-case failure above is about the simplex draw. One might hope to repair it, for example by placing mass at $\theta^\kappa = 0$, or by treating seeds as the only permitted base case. The second obstruction shows why such repairs do not restore Eq. equation 5.

Proposition 2 (No edge-factorised structure). *Under the process of equation 2–equation 3, the event “ v is ever positively activated” is not a function of a per-edge independent random structure. Consequently no live-edge graph and no trigger set exists whose reachability relation reproduces the process.*

Proof (Proof sketch). By Observation 2, after conditioning on the window draw the entire trajectory — including which nodes are ever positive and the final spread — is a deterministic function of S . The only random object is the window draw, which is indexed by *node*, not by edge, and which does not fail independently per edge. The activation rule equation 3 depends on the *sum* $\delta(v, t)$ of weights over a dynamically growing set $A_t^{\text{in}}(v)$; it is therefore not expressible as “at least one incident edge succeeded” (the IC form) nor as “the set of active in-neighbours covers a sampled threshold set” with a per-edge independent

threshold (the LT form). Since the reachability relation in Eq. equation 5 must be induced by such a structure, and no such structure exists, the identity is unavailable.

Two consequences deserve emphasis.

- **This is a domain problem, not an efficiency problem.** The RR family — TIM, TIM+, IMM, OPIM, SSA, D-SSA — and its GPU-parallel descendants all consume the same identity. None of them can be instantiated, so their asymptotic guarantees and their orders-of-magnitude speedups do not transfer. There is no RR baseline to beat in this setting.
- **The result does not depend on non-monotonicity.** Even if g were replaced by a monotone function, Proposition 2 would still hold, because it concerns the factorisation of the randomness rather than the shape of the objective. This matters because the objective in this setting turns out to be monotone over most of the tested range (§5), so an argument phrased purely in terms of non-monotonicity would be fragile.

4.3 The source model avoids the objective, and we can say how

It is worth recording that the model’s own solution path is consistent with the above. Its analysis establishes only a γ/k ratio for incremental greedy — not $(1 - 1/e)$ — and its upper-bound method is constructed by exhibiting two *monotone submodular* functions $\sigma^\kappa(\cdot)$ and $\sigma^\tau(\cdot)$ derived from linear-threshold processes, with $\lambda(S) \geq \sigma(S)$, and then optimising the surrogate λ . Since σ^κ and σ^τ are genuine LT influence functions, RR sampling *can* be applied to them. The correct statement is therefore not “RR is impossible in this problem” but:

RR can only be applied to a monotone upper bound $\lambda(S) \geq \sigma(S)$, whose optimisation guarantees a γ/k ratio at best; applying coverage semantics directly to σ makes the estimator degenerate to zero.

Quantifying the surrogate gap $\lambda(S) - \sigma(S)$ is a natural next step and we flag it in §8.

5 The Controlling Variable Is the Seed Fraction

5.1 Why absolute budgets mislead

Our first two attempts to characterise this problem produced the conclusion that out-degree *does not* fail under overexposure, and both were wrong for the same reason. We sampled candidate nodes uniformly from the whole graph and correlated out-degree with the true marginal gain. On a heavy-tailed graph such a pool consists almost entirely of low-degree nodes, so the correlation is dominated by the global regularity “high-degree nodes are more valuable” rather than by any ability to rank candidates under a given state. Worse, the experiments were

Table 2. Out-degree becomes an anti-signal once the network saturates. Values are means over $|S|/n \geq 20\%$. “neg.” is the share of candidates with $\Delta(v \mid S) < 0$. δ_2 denotes the state-conditioned closed form of §6. Trust Bitcoin-Alpha is one of the source model’s own datasets.

graph	n	$\langle k \rangle$	ρ_{degree}	ρ_{δ_2}	neg.
NetHEPT	15233	4.23	+0.092	+0.686	0.0%
p2p-Gnutella08	6301	6.59	−0.094	+0.756	8.0%
ca-GrQc	5242	11.06	−0.701	+0.656	20.0%
Trust Bitcoin-Alpha	3783	12.79	−0.679	+0.592	10.8%
Wiki-Vote	7115	29.15	−0.035	+0.265	8.0%
ca-HepPh	12008	39.48	−0.431	+0.481	21.0%
Facebook	4039	43.69	−0.402	+0.388	21.0%
email-Eu-core	1005	50.89	−0.695	+0.685	34.0%
Congress-Twitter	475	55.95	−0.049	+0.050	35.0%

run at tiny coverage: on NetHEPT ($n = 15233$) a budget of 200 seeds is 1.3% of the network.

Replacing the uniform pool by a degree-retrieved pool — which is what a deployed *retrieve-then-rerank* pipeline actually sees — moves the measured out-degree rank correlation from 0.882 to 0.654 on the same graph and objective, because the pool no longer spans the degree distribution.

The decisive variable, however, is not the pool but the *seed fraction* $|S|/n$. Saturation is a property of coverage, not of the absolute budget. The same budget of 200 seeds is 1.3% of NetHEPT but 42% of Congress-Twitter ($n = 475$).

5.2 Sweeping the fraction across nine graphs

We therefore sweep $|S|/n \in \{0, 0.05, 0.10, 0.20, 0.40\}$ on nine graphs whose mean degree spans 4.23 to 55.95, and measure the rank correlation between out-degree and the true marginal gain $\Delta(v \mid S) = \sigma(S \cup \{v\}) - \sigma(S)$, estimated by paired Monte Carlo with windows shared across candidates within a trial.

Three observations follow from Table 2.

1. **Out-degree flips sign.** It is positively correlated with the marginal gain on only one of nine graphs — NetHEPT, the sparsest — and negative on the other eight once $|S|/n \geq 20\%$.
2. **The objective really is non-monotone there.** Between 8% and 35% of candidates have a *negative* marginal gain on the denser graphs. This matches the source model’s own report that the objective value fluctuates with the budget on its sparsest dataset.
3. **The state-conditioned closed form stays positive on all nine.** It is the only scorer we tested that does.

Density is not the explanation. It is tempting to attribute the sign flip to density. We do not, because the evidence does not support it: with nine graphs the

rank correlation between $\langle k \rangle$ and ρ_{degree} is only -0.183 — weaker than the -0.405 we measured with eight, so adding a graph moved it *towards* zero — and Congress-Twitter, the densest graph, is a counterexample at $\rho_{\text{degree}} = -0.049$. The defensible claim is about the *saturated regime*, not about dense graphs.

A structural hint. After in-weight normalisation, exposure δ is driven by a node’s *in*-degree while a seed’s value is driven by its *out*-degree. The two coincide only when a graph is approximately symmetric. Consistently, the graphs with $\rho(\text{in}, \text{out})$ near 1 (ca-GrQc and ca-HepPh, both symmetrised collaboration networks, and Trust Bitcoin-Alpha at $+0.862$) show strongly negative out-degree correlations, while NetHEPT, whose in- and out-degrees are nearly independent ($\rho = -0.051$), is the one graph where degree survives. We record this as a hypothesis rather than a result; testing it needs more graphs than the nine available here.

6 A Training-Free State-Conditioned Score

6.1 Definition

We need a scorer that (i) requires no sampling, and (ii) reads the *realised* exposure state. Observation 2 makes the state cheap to obtain: given the windows, one cascade returns $\delta(v)$ for every v , at a cost of one simulation per decision rather than one per candidate.

Let $g(e) = 2e(1 - e)$ by Remark 1, and let $N^+(w)$ be the out-neighbours of a candidate w , each with weight ω_{wv} . Adding w raises the exposure of each $v \in N^+(w)$ by

$$\Delta e_w(v) = \omega_{wv} (1 - \delta(v)), \quad (6)$$

the factor $(1 - \delta(v))$ accounting for the influence v has already accumulated. The score of w is the resulting change in the expected number of its positively active out-neighbours,

$$\text{score}(w) = \sum_{v \in N^+(w)} \left[g(\delta(v) + \Delta e_w(v)) - g(\delta(v)) \right]. \quad (7)$$

We take two hops: equation 6 is propagated once more through the out-edges of each v , each hop attenuated by $(1 - \delta(\cdot))$, before equation 7 is evaluated. We refer to equation 7 as δ_2 .

Two properties matter for what follows. First, equation 7 is a *closed form*: no Monte-Carlo trials are involved, so its decisions carry no estimation noise. Second, because g is concave and non-monotone, equation 7 can be *negative* — it penalises a candidate that would push already-exposed neighbours past their overexposure threshold. A negative score expresses “do not seed this”, which is exactly the information that is absent from a degree ranking.

Table 3. Calibration of δ_2 against the true marginal (Congress-Twitter, 40 candidates, 60 paired trials per candidate). The score is a good *ranker* and a poor *calibrator*: the ratio is far from 1 and its sign is unstable, while the rank correlation stays positive throughout.

$ S /n$	true mean marginal	δ_2	mean	ratio	Spearman
0%	+193.586	+2.852	0.01	0.902	
5%	-0.661	-0.251	0.38	0.667	
10%	+0.167	-0.095	-0.57	0.263	
20%	+0.120	-0.192	-1.60	0.427	

6.2 Why a closed form can beat a noisy exact method

equation 7 is not an unbiased estimator of $\Delta(w \mid S)$; empirically it underestimates the marginal by roughly two orders of magnitude at $|S|/n = 0$ and its sign can differ from the truth once saturated. That sounds disqualifying and is not, for a reason that is central to this paper.

Once the network saturates, the true marginal $\Delta(w \mid S)$ is close to zero for *every* candidate. The spread of the true marginals across candidates therefore also collapses. A method that estimates the marginal by Monte Carlo — which is what exact greedy does — then faces a signal-to-noise problem: the differences it must resolve are smaller than its own estimation error, so its argmax is driven by noise. A method that is *biased but noise-free* can still rank usefully, and — as §7 shows — does.

This is not a claim that bias is harmless. It is a claim about which error dominates in this specific regime, and we quantify it in §7.

The practical consequence, visible in Table 3: the *zero crossing* of δ_2 is not an unbiased estimate of the point at which adding seeds stops helping. An early version of our pipeline used “stop when score ≤ 0 ” as a stopping rule and halted after a single seed, scoring 347 against a degree baseline of 364. Stopping decisions must therefore come from the observed spread — which is exact, by Observation 2 — rather than from the analytic score. We use a patience-based stop on the observed spread in §7.

7 Experiments

7.1 Setup

Graphs. Table 2 lists the nine graphs. All are loaded with in-edge weights normalised so that $\sum_u \omega_{uv} = 1$ for every v with in-edges. This normalisation is not a convenience: the model *requires* weights large enough that a plausible number of active in-neighbours can push δ into the window. With uniform weights $p = 0.01$ on a sparse graph, δ is bounded by $0.01 \times \text{in-degree}$ — at most 0.6 and in practice about 0.15 on NetHEPT — so the cascade barely fires (mean marginal gain 1.046, i.e. essentially the seed itself) and no overexposure is observable at all. The datasets used by the source model already carry such normalised weights.

Table 4. Label signal-to-noise in the negative-marginal regime (Congress-Twitter, 30 candidates, paired labels). At 60 trials the noise equals the signal; only at ≥ 300 trials are rank correlations interpretable.

$ S /n$	MC	mean	gain	sd(gain)	mean	se	SNR	ρ_{δ_2}
10%	60	+0.084	0.514	0.525	0.98	+0.010		
10%	250	−0.245	0.338	0.249	1.36	+0.588		
10%	800	−0.179	0.349	0.132	2.65	+0.613		
20%	60	−0.018	0.521	0.383	1.36	+0.469		
20%	800	−0.106	0.268	0.096	2.79	+0.828		

Protocol. Two design choices matter and we state them explicitly because ignoring either produces wrong conclusions.

1. **Measure the seed fraction, not the budget.** As §5 shows, the regime is determined by $|S|/n$. Every table reports both.
2. **Use paired evaluation with shared random numbers.** We compare policies by running them on *the same* threshold windows and reporting the difference in spread. Because the process is deterministic given the windows (Observation 2), this cancels the dominant source of variance and resolves marginal differences that an unpaired comparison cannot.

7.2 A measurement caveat that constrains the field

Before reporting ranking results we must report a trap we fell into. In the regime where the true marginal is near zero, the *labels themselves* are hard to measure. Table 4 reports the signal-to-noise ratio of paired marginal-gain labels on Congress-Twitter: the ratio of the spread of the true marginals to the mean standard error of their estimates.

At 60 trials the SNR is below 1.4: label noise is as large as the dispersion being measured. An earlier version of our own analysis concluded from such numbers that “all scorers fail once saturated”; at 800 trials the same measurement gives $\rho_{\delta_2} = 0.61$ – 0.83 . We therefore restrict rank-correlation claims in this regime to $MC \geq 300$, and prefer paired spread comparisons, which sidestep the issue entirely. We report this because it is actionable for anyone working on overexposure, and because we have not seen it stated.

7.3 Main comparison

Table 5 compares three training-free policies on Congress-Twitter, at a budget of 10 additional seeds drawn from a pool of 60, over 400 paired trials.

Unsaturated, everything is equal. At $|S|/n = 0$ the three policies are within 1.5% of one another. Degree is already adequate, and there is nothing to improve. This is worth stating plainly: the interesting regime is $|S|/n \geq 5\%$, and work that reports only low-coverage results will find no effect.

Table 5. Congress-Twitter, budget 10, pool 60, 400 paired trials with per-trial shared windows. The quantity is the marginal spread added by the budget over the pre-existing seed set. Degree becomes harmful once saturated; the closed form δ_2 beats both degree and exact Monte-Carlo greedy.

$ S $	$ S /n$	degree top-10	δ_2 top-10	exact MC-greedy top-10
0	0.0%	$+362.51 \pm 0.61$	$+361.64 \pm 0.59$	$+367.28 \pm 0.58$
24	5.1%	-4.39 ± 0.48	$+0.49 \pm 0.39$	-0.01 ± 0.45
48	10.1%	-1.61 ± 0.47	$+2.24 \pm 0.32$	$+0.28 \pm 0.46$
95	20.0%	-5.17 ± 0.40	$+2.26 \pm 0.24$	$+0.42 \pm 0.31$

Saturated, degree is harmful. Adding the ten highest-degree nodes *reduces* the spread by 1.6 to 5.2 relative to the pre-existing seed set. At $|S|/n = 20\%$ the deficit is -5.17 ± 0.40 , i.e. 13 standard errors. A degree heuristic is not merely suboptimal here; it is worse than doing nothing.

Exact greedy also fails, and the closed form wins. Exact Monte-Carlo greedy recovers only $+0.42 \pm 0.31$ at $|S|/n = 20\%$, whereas δ_2 recovers $+2.26 \pm 0.24$ — more than 4 standard errors better. The mechanism is the one given in §6: the greedy searches for the largest marginal using $MC = 25$ estimates while the true marginals are all near zero, so its selection is noise-driven. The closed form, being deterministic, is not.

7.4 Comparison against the source model’s own algorithms

The source model evaluates its methods against **Random**, **Max_Degree**, **IMRank**, **PageRank** and **CELF**. Since we are questioning the machinery those methods avoid, we implement all five, together with the two algorithms the source model actually proposes: incremental greedy (IGA, its Algorithm 1), which greedily maximises the objective, and its upper-bound method (UB), which optimises the surrogate $\lambda(S) = \sigma^\kappa(S) - \sigma^\tau(S)$. Every policy is evaluated on the true objective under a paired protocol with shared threshold windows.

Four points from Table 6.

The proposed method does not beat a zero-cost closed form. IGA attains a mean marginal of $+132.21$ against δ_2 ’s $+133.20$, while spending 13,875 cascade simulations per seed set against δ_2 ’s zero. Its exact variant CELF reaches $+132.86$ at the same cost and still falls short. The gap widens sharply in the regime this paper is about: restricted to $|S|/n \geq 10\%$, δ_2 averages $+8.13$ while IGA and CELF average $+4.66$ and $+4.74$ — roughly half, for three orders of magnitude more computation. On Congress-Twitter at $|S|/n = 10\%$ both are *negative* (-1.33 ± 0.80 and -0.31 ± 0.71) where δ_2 reaches $+6.20 \pm 0.16$. The γ/k guarantee the source model proves for IGA is consistent with this: it is a weak bound, and empirically the method is the wrong tool once the network saturates.

Table 6. All policies evaluated on the true objective. Marginal spread over the pre-existing seed set, over three graphs $\times$ three seed fractions ($n = 9$ cells, 150 paired trials each). “satur.” restricts to $|S|/n \geq 10\%$. **casc.** is the number of cascade simulations the policy spends per query. UB arms implement our reading of the stated definition — see §8.

policy	mean	satur.	worst case	casc.
δ_2 (this paper)	+ 133.20	+ 8.13	+ 1.17	0
CELF [4]	+132.86	+4.74	−0.48	13,875
IGA [13]	+132.21	+4.66	−1.33	13,875
degree (Max_Degree)	+129.50	+1.84	−3.85	0
IMRANK	+126.05	+3.52	−0.55	0
Random	+123.41	+2.75	−1.90	0
PageRank	+122.76	+2.89	+0.24	0
UB- κ	+107.30	+5.89	−0.63	0
UB- λ	+104.92	+3.47	−0.63	0

δ_2 is the only policy whose worst case is comfortably positive. Every other row has a negative cell in the sweep; **PageRank** is the sole baseline that never goes negative, at +0.24, but it is second-worst overall.

Optimising the surrogate produces a poor seed set overall. The UB arms are the two weakest policies, despite σ^κ and σ^τ being genuine monotone submodular influence functions. The mechanics are visible: on Congress-Twitter a single seed already activates 467 of 475 nodes under the lower threshold θ^κ , so greedy on the surrogate stops after one or three seeds because no further seed has positive marginal gain *under the surrogate* — while those same seeds leave substantial spread unrealised under the true objective. Notably UB- κ does comparatively well *inside* the saturated subset (+5.89, second only to δ_2), so the surrogate is not uniformly useless; it is the low-coverage behaviour and the early termination that hurt. This is the practical face of the gap measured in §8.

The ordering is not explained by cost. **PageRank**, **Random** and IMRANK all spend zero cascades and beat both UB arms, which also spend zero. What separates the rows is whether a policy reacts to the overexposure *state*: only δ_2 does, and it is the only one that is never harmful.

7.5 Robustness across selection draws

A single selection draw could make δ_2 look good by luck. We therefore repeated the whole select-then-evaluate procedure over 8 independent draws, scoring every selection on a shared set of 100 evaluation windows so that only the selection varies.

The degree baseline is itself stable across draws (+5.07 to +5.97 on the 10% block), which confirms that the degradation it suffers is real rather than a selection artefact.

Table 7. Robustness across 8 independent selection draws (Congress-Twitter, budget 10, pool 60). δ_2 beats degree on every draw at every fraction; no counterexample exists.

$ S /n$	degree	δ_2 mean [min, max]	wins
0%	+362.72	+366.30 [364.29, 368.05]	8/8
10%	-2.51	+1.19 [0.48, 2.10]	8/8
20%	-5.55	+0.23 [0.09, 0.42]	8/8

Table 8. NetHEPT ($\langle k \rangle = 4.23$). Degree remains beneficial at every fraction and actually wins at $|S|/n = 20\%$.

$ S /n$	degree	δ_2
0%	+123.45 $\pm$ 5.53	+126.21 $\pm$ 5.58
10%	+41.80 $\pm$ 2.07	+49.05 $\pm$ 2.00
20%	+15.70 $\pm$ 0.94	+11.05 $\pm$ 0.92
40%	+7.04 $\pm$ 0.40	+9.30 $\pm$ 0.65

Stopping. Because δ_2 is not calibrated (Table 3), we pair it with a *patience* stop on the observed spread rather than on the score. With patience 2 the policy stops after 3 to 6 seeds and still converts the degree deficit into a gain at every saturated fraction (5%: $-4.59 \rightarrow -0.50$; 10%: $-1.97 \rightarrow +2.23$; 20%: $-5.06 \rightarrow +1.00$ over 200 paired trials), spending one cascade per step.

7.6 The advantage is graph-dependent

On NetHEPT the picture is different, and we report it because it bounds the claim.

On the sparsest graph, saturation is reached through the *number* of seeds rather than through any individual node’s degree, so degree does not become an anti-signal. The correct claim is therefore “in the saturated regime, on graphs where it bites”, not “degree is universally harmful”.

8 Limitations and Threats to Validity

We list the things that could be wrong, in the order we consider them most likely to matter.

1. *A learned scorer does not beat the closed form.* Having established that δ_2 beats degree and exact greedy, the natural next question is whether a learned state-conditioned scorer can beat δ_2 . On the evidence we collected, it cannot. We built a dataset of 480 labelled states on Congress-Twitter (4 seed fractions $\times$ 3 independent contexts $\times$ 40 candidates, paired labels at MC = 400) and evaluated with *leave-one-context-out*, so every score is a cross-state generalisation score. With $\text{regret} = 1 - (\text{model top-5 gain})/(\text{true top-5 gain})$:

model	Spearman	regret
δ_2 (no training)	+0.6932	0.5025
ridge on the same features	+0.4255	0.6172
MLP on the same features	+0.3652	0.9662

A feature ablation, replicated on two independently generated datasets, shows that this is not a matter of insufficient model capacity. Adding threshold-derived features (distance to θ^τ , fraction of neighbours inside the band) to the analytic score makes held-out regret *worse* on both replications (+0.167 and +0.300), and on the first replication the combination wins 0/12 held-out contexts. The single analytic score column is the strongest single predictor we found.

This should be read as a limitation, not as a theorem: 480 states is small, and we cannot exclude that a larger corpus would change the answer. What we can say is that under a cross-state evaluation protocol, at this scale, the closed form is not beaten, and the threshold features — which were our main hypothesis for where extra signal would come from — are actively unhelpful. We report it because a paper that shows a strong training-free baseline should also report that learning did not improve on it.

2. A trap in interpreting feature importance here. Pooled across contexts, *every* informative feature correlates with the label at $|r| \approx 0.85\text{--}0.91$, including one at 0.907 that is pure regime identification. This is because the dataset mixes regimes whose label scales differ by three orders of magnitude ($\approx +200$ when unsaturated, ≈ 0 when saturated). Any evaluation that does not hold out a context will therefore report strong, meaningless feature utility. We mention it because we initially drew exactly that wrong conclusion.

3. Density does not explain the sign flip. We would like a crisp graph-level criterion predicting when out-degree fails. We do not have one. Across the nine graphs the correlation between $\langle k \rangle$ and ρ_{degree} is -0.183 , and Congress-Twitter is a counterexample — indeed the correlation moved *towards* zero when the ninth graph was added, which is the opposite of what a density-based criterion would predict. The structural hint in §5 (in-/out-degree coupling) is a hypothesis supported by a $\rho(\text{in}, \text{out}) = 1.000$ symmetry in the two collaboration graphs, but nine graphs is not enough to establish it.

4. Scale and dataset coverage. Our largest graph has 15,233 nodes; the source model’s datasets are smaller still. We do not demonstrate behaviour on million-node graphs, and the per-decision cost of obtaining the state — one cascade, which is cheap here — would need re-examination at that scale.

5. The surrogate gap is measured, and it exposes an open question. §4 observes that RR is applicable to the monotone upper bound $\lambda(S) \geq \sigma(S)$ used by the source model. We therefore measured the gap $\lambda(S) - \sigma(S)$ directly, which the source model does not report (Table 9).

Table 9. The surrogate gap. $\lambda = \sigma^\kappa - \sigma^\tau$ where $\sigma^\kappa, \sigma^\tau$ are linear-threshold spreads at the lower and upper thresholds. All quantities share the same window draw within a trial.

graph	$ S /n$	$\mathbb{E}[A]$	λ	gap	gap/ λ
Congress-Twitter	5.1%	367.17	441.40	+74.23	+0.170
Congress-Twitter	10.1%	366.13	403.53	+37.40	+0.098
Congress-Twitter	20.0%	363.57	342.23	-21.33	-0.065
NetHEPT	5.0%	1469.87	1442.00	-27.87	-0.019
NetHEPT	10.0%	2787.77	2329.97	-457.80	-0.197
NetHEPT	20.0%	4672.73	2387.00	-2285.73	-0.958

Two things follow. First, the bound is *tight where it is useful*: on Congress-Twitter at $|S|/n = 10\%$ it over-estimates the objective by 9.8%, which makes the surrogate worth optimising — and σ^κ and σ^τ are genuine LT influence functions, so RR sampling applies to each individually. Second, the gap degrades with saturation and, under our reading of the definitions, *inverts*: $\lambda < \sigma$ in 4 of 6 settings. Across a wider sweep (4 graphs $\times$ 3 fractions $\times$ 30 trials) the per-realisation containment that the source model’s proof invokes holds in 0/180 draws and the inequality $|A| \leq |A^\kappa| - |A^\tau|$ holds in 73/180.

We deliberately do not present this as a refutation. The source states Theorem 6 for the expectation of σ but proves it on one fixed realisation, and at least three innocent readings remain open: our interpretation of $\sigma^\kappa/\sigma^\tau$ may differ from the intended one, the two threshold processes may be intended to be coupled rather than independent, or the proof may be illustrative rather than general. Every counterexample we observe is a node active in the θ^τ process but not the θ^κ process, which is exactly what a monotone coupling would forbid. Resolving this requires clarification from the authors, and we flag it as an open question rather than a result. What we can state without qualification is the measured gap above, and the structural point that follows: λ is a *difference* of two monotone submodular functions, and submodularity is not preserved under differences, so even a valid upper bound would not by itself license a sampling-based guarantee.

6. Non-monotonicity is regime-dependent, and we once got this wrong. We initially reported that the objective is effectively monotone, based on a sweep at $|S| \leq 4000$ on a graph where that is at most 26% coverage and mostly far less. Measured as a *fraction*, the objective is clearly non-monotone on most of our graphs (Table 2). Any statement about overexposure that does not report $|S|/n$ should be treated with suspicion — including, historically, ours.

9 Conclusion

We set out to select seeds under a threshold-dependent overexposure model and found that the dominant algorithmic tool for influence maximization cannot be used at all. Reverse-reachability sampling requires a randomised per-edge

structure and a base case in which a node is active under incoming influence from a single predecessor. Under the threshold-window process neither exists: the activation threshold is strictly positive almost surely, so zero-influence nodes are never active and every reverse walk terminates immediately; and the exposure score is a set function of the ever-activated in-neighbours, so after conditioning on the windows the whole trajectory is deterministic in the seed set. Measured directly, every RR set is empty and the classical estimator returns 0 where the true spread is 111.9. There is no RR baseline to beat here, and the results that make RR attractive — near-linear time with $(1 - 1/e - \varepsilon)$ guarantees, and the GPU speedups built on it — do not transfer.

What replaces it is unglamorous but effective. The regime that matters is the seed *fraction*: across nine graphs the objective becomes genuinely non-monotone once $|S|/n \geq 20\%$, with 8% to 35% of candidates having a negative marginal gain, and out-degree turning from a signal into an anti-signal on eight of the nine. Exact Monte-Carlo greedy also degrades there, because the marginals it must resolve are smaller than its own estimation noise. A two-hop closed form that reads the realised exposure state and penalises candidates that would over-expose their neighbourhood recovers spread where degree destroys it, beats exact greedy, and does so on 8/8 independent selection draws — at one cascade per decision rather than $k \cdot |M| \cdot \text{MC}$.

We close on a negative result rather than a new method. Having found a strong training-free baseline, we asked whether a learned state-conditioned scorer could beat it, and under a cross-state evaluation protocol it could not: the closed form’s regret against the true top-5 is 0.34–0.50, and every learned variant we tried was worse, with our main hypothesis for where extra signal would come from — the threshold information the closed form ignores — turning out to be actively harmful. A remaining gap of that size means the problem is not solved; it means the obvious learned approach is not the way to close it.

Two directions follow directly. The first is to quantify the surrogate gap $\lambda(S) - \sigma(S)$ between the objective and the monotone upper bound to which RR *can* be applied. We took a first step: the bound is tight at low coverage (9.8% over-estimate on Congress-Twitter at $|S|/n = 10\%$) but degrades with saturation, and under our reading of the definitions it inverts, which raises a question about the expectation-versus- realisation step in its proof that we flag rather than resolve. Because λ is a difference of two monotone submodular functions and differences do not preserve submodularity, even a valid upper bound would not by itself license a sampling-based guarantee. The second direction is to look for a graph-level criterion — we suspect in-/out-degree coupling — that predicts when degree-based heuristics fail, which would let a practitioner decide whether the overexposure regime needs to be handled at all.

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